BIKE RENTAL PREDICTION

Course-end Project 2 | Random Forest Regression

Project write-up prepared from the supplied project brief and dataset (dataset.xlsx, sheet “day”).

1. Introduction

Bike-sharing systems allow customers to rent and return bicycles through an automated service. Rental demand can vary with weather, season, month, holidays and working-day patterns. Understanding these relationships can help a bike rental company plan bicycle availability and daily operations.

2. Problem Statement

The objective of this project is to analyze daily bike rental data and build a machine-learning model that predicts the total number of bikes rented (cnt) from environmental, seasonal and calendar-related variables.

3. Dataset Description

The supplied Excel workbook contains a worksheet named “day”. It has 731 daily records and 16 columns, covering 1 January 2011 through 31 December 2012.

4. Exploratory Data Analysis

4.1 Data types

The date field (dteday) is converted to a Date type. Season, year, month, holiday, weekday, workingday and weathersit are treated as categorical variables. The target cnt remains numeric for regression.

4.2 Missing values

The uploaded dataset contains 0 missing values across all 16 columns. Therefore, no missing-value imputation is required for this dataset.

4.3 Rental-count summary

The total daily rental count (cnt) ranges from 22 to 8,714. The mean is approximately 4,504 rentals per day, and the median is 4,548.

4.4 Monthly trend

When the two years are combined, June has the highest average daily rental count (approximately 5,772), while January has the lowest (approximately 2,176). The overall pattern indicates that rentals tend to rise from winter into warmer months, although demand varies across the year.

4.5 Yearly trend

The average daily rental count is approximately 3,406 in 2011 and 5,600 in 2012. This indicates higher average daily usage in 2012 than in 2011.

4.6 Outlier analysis

Boxplots are generated for cnt and selected weather variables. Values beyond the boxplot whiskers are flagged as potential outliers. They should be investigated rather than automatically deleted, because high rental counts may represent genuine peak-demand days.

5. Data Preparation

The data is split into 80% training and 20% testing subsets using a fixed random seed for reproducibility. The response variable is cnt. The record index instant and raw date dteday are excluded from model predictors.

The columns casual and registered are also excluded because cnt is the sum of these two counts. Including them would introduce target leakage and would not represent a valid prediction made before the rentals are known.

6. Model Building: Random Forest Regression

A Random Forest regression model is trained using 500 trees. Random Forest combines predictions from multiple decision trees and can model nonlinear relationships and interactions among predictor variables. The model uses season, year, month, holiday, weekday, workingday, weather situation, temperature, feeling temperature, humidity and wind speed as predictors.

7. Model Evaluation

Predictions are generated for the held-out test dataset. The R script calculates the following metrics. Run the script and enter the resulting values in the table before submitting the report.

An actual-versus-predicted scatter plot is also produced. Points closer to the diagonal reference line indicate predictions closer to the observed rental counts. Variable-importance output is included to compare the predictive contribution of the input features.

8. Conclusion

This project demonstrates an end-to-end regression workflow for bike rental demand: data import, type conversion, missing-value checks, visualization, train-test splitting, Random Forest model fitting and test-set evaluation. Exploratory analysis shows clear seasonal variation and higher average daily rental counts in 2012 than in 2011. The final model's predictive performance should be reported using the RMSE, MAE and R-squared values printed by the R script. These metrics should be added to this report after execution.

9. Files and Submission Checklist

- Keep Bike_Rental_Prediction.R and dataset.xlsx in the same working folder.
- Open the R script in RStudio and run it from top to bottom.
- If required, install the packages listed at the top of the R script.
- Capture screenshots of the code and outputs/plots as requested in the course project brief.
- Replace the three [Enter output] placeholders with the actual model metrics.
- Submit the R script and completed write-up with the required screenshots according to the course instructions.

Note: Model evaluation values are intentionally left blank in this write-up because they must be generated by executing the R model. No model-performance numbers have been fabricated.

Tables from the project write-up

Variable group	Columns	Purpose
Record/date	instant, dteday	Record index and date
Calendar and season	season, yr, mnth, holiday, weekday, workingday	Represent seasonal and calendar conditions
Weather	weathersit, temp, atemp, hum, windspeed	Represent weather and environmental conditions
User counts / target	casual, registered, cnt	User counts; cnt is the target variable

Metric	Meaning	Result from R run
RMSE	Root mean squared error; penalizes larger prediction errors	[Enter output]
MAE	Mean absolute error; average absolute error in bike counts	[Enter output]
R-squared	Proportion of test-set variation explained by predictions	[Enter output]